SpeakSmart: A Comprehensive Oratory Enhancement Tool

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Abstract. Communication skills are crucial for success in many fields and industries, especially those requiring public speaking. However, many people find public speaking to be a challenging and anxiety-inducing task. To address this, we aim to develop a user-friendly personalised tool to help individuals with their public speaking skills by utilising computer vision and speech recognition technologies. The purpose of this study is to judge an individual's public speaking abilities by analysing video footage of the individual. The analysis focuses on three main elements: voice, speech and body language of the user. Features such as facial expressions, hand gestures, posture, and auditory parameters like disfluencies are extracted and assessed. Individuals can thus receive insights on their skills tailored to their needs and use them to enhance their public speaking and build confidence in their speech over time.

Keywords: Grammar Error Correction, Disfluency Detection, Affective Computing, Sentiment Analysis, Posture Analysis, Hand Gesture Analysis.

1 Introduction

Oratory, the art and practice of formal speaking, is an essential skill in both professional and personal contexts, particularly when addressing an audience. The ability of delivering compelling public speeches is important to many, as it allows presenting information convincingly and powerfully. On many different occasions, including delivering educational, professional and informative sessions and so on, having strong skills in public speaking grants the speaker the power for persuasion and generating ideas. While the content of a speech is essential, its delivery, including body language, vocal modulation, and nonverbal cues, plays an equally important role in determining the impact of a message. Non-verbal cues such as facial expressions, gestures, posture and eye contact have a major impact on how your message is received and the audience's level of interest. Similarly, auditory features such as volume, pitch, frequency, pauses, repetition and the use of filler words such as "umm", "ah", "for example", "that is to

say" and "you see" are important factors in determining a speaker's confidence and clarity. Collectively, these spoken and unspoken components influence the overall impact of public speaking.

In this setting, we suggest creating a thorough, easy-to-use tool that utilizes computer vision and speech recognition to evaluate a person's public speaking abilities. This system is intended to assess two essential aspects of an oral presentation: nonverbal communication and verbal delivery. The objective is to judge an individual's public speaking abilities by analysing audio and video footage of the individual. After the user submits a video recording of themselves talking, making sure their upper body is visible, multimodal analysis shall cover various facets of the presentation. The auditory aspects encompass speech disfluencies, emotional tone, anxiety, and tension. Visual parameters emphasize facial expressions, hand movements, and body posture. Moreover, textual parameters assess the grammatical correctness of the transcript and detect underlying textual sentiments. By combining these multi-faceted evaluations, the tool seeks to deliver comprehensive feedback that enables users to improve their speaking abilities successfully.

The structure of the remainder of this paper is as follows: Section 2 accommodates literature survey covering existing approaches to the mentioned methods, focusing on recent advances in all the domains. Section 3 outlines methodology, including the architecture of the proposed system. The implementation details, covering dataset, preprocessing and model training are discussed in Section 4. Section 5 shows the results of the experiments, followed by an analysis of the system's performance. Finally, Section 6 talks about the conclusion of the paper and potential discussions in directions for future work.

2 Previous Work

Numerous approaches have been used to investigate facial emotion recognition. Using eye-tracking for emotion analysis, Lim et al. [1] used ANNs to classify with 90% accuracy in two seconds. However, the study highlighted a critical limitation in its inability to account for cultural variations in emotional expression.

Mustaqeem et al. [2] used a BiLSTM model with RBF-based K-means clustering in speech emotion recognition (SER), attaining 85.57% accuracy on EMO-DB and 77.02% on RAVDESS. However, the method faced high computational costs and reliance on dataset quality, as reflected in the performance differences between datasets. Using multilingual datasets, Wani et al. [3] examined SER approaches and discovered that SVMs performed better than other classifiers, with an accuracy of 89.2%. Although it had trouble with cultural dependencies, data preprocessing (DFT framing, normalization, and MMSE noise reduction) enhanced dataset quality. Using an RBF SVM on RML, another study [4] concentrated on audio feature-based emotion categorization (MFCC, ZCR, HNR, and TEO). While effective, this approach was computationally intensive and suffered from emotion-context dependency, posing challenges for broader applications.

Kourkounakis et al. [5] introduced FluentNet (SE-ResNet + BiLSTMs) to classify disfluencies. STFT (Short Time Fourier Transform) was used to obtain spectrograms of 4s audio clips to feed into the model, resulting in a balanced dataset (LibriStutter - 20 hours, 18,000 samples). The model achieved 91.75% accuracy on UCLASS and 86.70% on LibriStutter. Mohapatra et al. [6] used wav2vec 2.0 [7] for creating contextual embeddings of the data, followed by CNN (DisfluencyNet) for classifying the various types of speech disfluencies labelled in the SEP-28k, obtaining 94.9% accuracy on "prolongation" and 75% on "interjections." Romana and Koishida [8] proposed a multimodal (wav2vec + BERT) approach for disfluency detection, improving ASR-based analysis. On the Switchboard dataset (a collection of about 2,400 two-sided telephone conversations among 543 speakers), manual transcriptions achieved 0.96 precision, 0.91 recall, 0.93 F1, while ASR yielded 0.86, 0.74, and 0.79, respectively.

For grammar correction, a BiLSTM-CNN hybrid model [9] combined PoS tagging and word embeddings, outperforming LSTM and CNN by 9% and 2% in accuracy, respectively, with an 8.2% F1-score increase. The study highlights the importance of pre-processing techniques, such as Byte Pair Encoding (BPE) and spaCy tokenization, which significantly contribute to improving the model's performance by facilitating efficient data representation and processing. While the paper does not explicitly state any limitations, the authors suggest future work focusing on enhancing the model's ability to detect sarcasm, which remains a challenging aspect of sentiment analysis. Hao and Hao [10] demonstrated Transformer models' effectiveness in GEC, detecting 25 error types using BPE and spaCy tokenization. Training on BEA-2019 ensured robustness. Reducing beam size from 12 to 1 significantly improved speed while maintaining competitive F0.5 scores (61.78 vs. 60.93). Larger beam sizes introduced computational overhead, highlighting optimization challenges. Chiche et al. [11] reviewed PoS tagging, finding 68% of solutions used DL (LSTM, RNN, BiLSTM), 20% ML, and 12% hybrid approaches. The review identified research gaps but was limited to studies before 2021.

3 Methodology

In order to evaluate verbal prowess and public speaking skills, the goal is to analyse all that may occlude a speaker's intended message - speech disfluencies, incorrect grammar, poor posture, lack of hand gestures and lack of emotional expression. The proposed methodology is to develop a tool, i.e. SpeakSmart (Fig. 1), that involves incorporating several independent modules for assessing each of the aforementioned criteria. Consequently, the expected input is video footage of a person speaking in plain view, which is preprocessed to extract voice, face and speech transcript. FFmpeg has been used to obtain the audio of the person's voice, which is then converted to a transcript using Whisper [12].

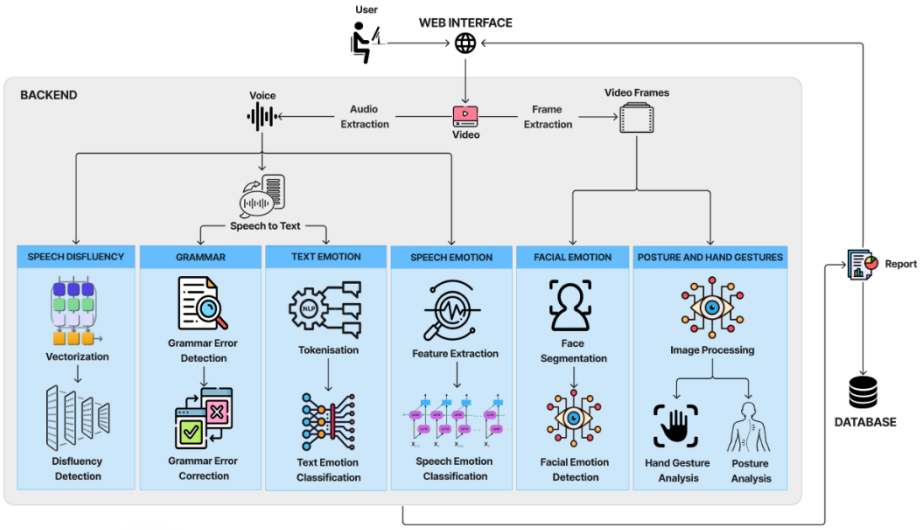


Fig. 1. - SpeakSmart

3.1 Grammar Correction

The grammar correction module works on the user's speech transcript, looking for any errors that may have crept in. Its working has been conceptualised as a three-phase process [13] as follows:

- Grammar error detection - Facebook's RoBERTa [14] for sequence classification
- Grammar error correction - Google's T5 [15] for sequence generation
- Error annotation using ERRANT [16]

Each of the aforementioned functions at the sentence level, with detection modelled as a binary classification task which determines whether a given sentence from the speech transcript has any grammatical errors. The advantage of RoBERTa lies in its deep contextual understanding, which helps it recognize both syntactic and semantic inconsistencies. The sentences that are flagged by the error detection model as incorrect are passed on to the second model, which is grammar error correction. This has been modelled as a sequence-to-sequence generation task specifically trained to correct targeted ten grammatical errors - Adjective Form, Adverb, Determiner, Conjunction, Preposition, Verb Form, Subject Verb Agreement, Verb Tense, Noun, Pronoun - where the incorrect sentence triggers the output of its grammatically correct equivalent. Finally, in order to gain perspective on the type of error that was present in the input, ERRANT, a rule-based annotator that works on parallel sentences, generates the error annotations corresponding to the incorrect-correct sentence pair.

3.2 Speech Disfluency

The disfluency module is audio-based, while looking into the following disfluencies - Interjection (filler words like "um", "like"), Prolongation (within a word), Block

(unnatural pause between words), Sound Repetition and Word Repetition. The audio extracted from the video is transformed into time series data using the torchaudio Python module. The detection occurs at every 3 seconds of the clip. In order to obtain a contextual vector representation of each 3 second clip, they are passed through wav2vec to obtain tensors of shape (1, 149, 768) [6]. To detect the presence of each disfluency, five separate CNN models are used to perform binary classification upon the embedding vector produced by wav2vec. Each CNN focuses on detecting one of the five types of disfluencies, trained to perform binary classification between fluent speech and the disfluency of interest. This approach ensures that each disfluency is analysed independently leading to a robust and precise detection system.

3.3 Facial Emotion

The facial emotion detection pipeline analyses the emotions from the video data employing a two-stage process that coalesces face detection and emotion classification. A YOLOv11 [17] model that has been fine-tuned for face detection, analyses and processes every frame of the video to identify bounding boxes corresponding to facial regions, which are then cropped for further examination. A second YOLOv11 model which has been fine-tuned for emotion classification receives these cropped facial regions, predicts emotion categories such as anger, neutral, fear, happy, sad, contempt, disgust and surprise. Each classified emotion is assigned a confidence score, allowing for the filtering of low-confidence predictions to enhance overall reliability. Video frames are processed at regular intervals for temporal consistency, and the detected emotions are aggregated to highlight the frequency of each emotion.

3.4 Hand Gesture & Posture

The posture and hand gesture detection module processes the input video data at the rate of 1 frame per second to classify body alignment, head positions, and hand posture. OpenCV's Perspective-n-Point (PnP) algorithm and MediaPipe [18] were utilized to estimate the head position by looking at the key facial landmarks identified by classifying orientations like "Looking Forward", "Looking Down", “Looking Left”, “Looking Right” based on the Euler angles calculated.
Hand positions are identified by comparing the wrist and shoulder landmarks, classifying gestures as "Hands Visible," "Hands Not Visible," or "Hands Folded". Postures such as "Standing Straight" or "Not Standing Straight." is classified by evaluating the shoulder alignment. A comprehensive frequency report offers insights about the subject's gestures and postures throughout the length of the video.

3.5 Speech Sentiment

The Speech Emotion Recognition Module processes audio from videos to classify emotions and evaluate the speaker's nervousness and stress levels. Using a Bidirectional Long Short-Term Memory (BiLSTM) model, it effectively captures long-term relationships in speech signals. The module analyses audio features in three-second intervals,

classifying emotions into six categories: Anger, Calm, Excited, Fear, Happy, and Sad. Features such as MFCCs, chroma, mel, contrast, tonnetz, pitch, jitter, shimmer, and intensity are extracted for classification. Nervousness is assessed by summing jitter and shimmer values, while stress is categorized as low, medium, or high based on thresholds for pitch, jitter, shimmer, and intensity. This approach ensures detailed and consistent emotional monitoring.

3.6 Text Sentiment

The Text Emotion Recognition module examines speech transcript sentences to detect emotional tones, offering insights into the speaker's emotional state through word choices and sentence structure. Using RoBERTa for sequence classification, it identifies six emotions: anger, surprise, fear, love, joy, and sadness. Text is tokenized with the RoBERTa tokenizer, converted into a suitable format, and adjusted to a uniform input size of up to 128 tokens. The optimized RoBERTa model then predicts the emotion with the highest score for each tokenized sentence. These emotions are recorded, providing a frequency-based overview of emotional trends in the transcript.

4 Implementation

4.1 Grammar Correction [13]

Dataset. The C4-200M dataset [19] is a large-scale corpus with 185 million English sentence pairs, generated using a Tagged Corruption model on Google's C4 dataset. Each pair includes a grammatically incorrect sentence and its corrected version.

Preprocessing. A sample of 90 million sentence pairs from the C4-200M dataset was processed using ERRANT to annotate errors. Only entries exhibiting specific errors of interest were retained, reducing the dataset to 12 million entries for focused training and evaluation. For binary classification, sentences in the pairs were labelled as 0 for grammatically incorrect and 1 for correct, creating a structured dataset for phase 1 of the GED task, which identifies if a sentence requires grammatical correction.

Grammar Error Detector. The error detector was fine-tuned for binary classification to determine whether input sequences are grammatically correct. The fine-tuned model includes 12 transformer layers leveraging self-attention for contextual understanding, followed by classification heads. Training used a learning rate of 5e-5, batch size of 64, and 3 epochs, optimised with AdamW for fast and stable convergence.

Grammar Error Corrector. The error corrector is based on Google's T5 model. It was fine-tuned for grammar correction, transforming incorrect sentences into their grammatically correct versions. Due to the dataset's size, it was split into chunks of 100,000 sentence pairs for efficient processing. Each chunk was loaded, tokenized, padded, and

passed to the model for training in batches. The AdamW optimizer with a learning rate of 5e-5 was used during training which ran for one epoch.

4.2 Speech Disfluency

Dataset. The dataset chosen for this task is Apple's SEP28K [20], a collection of 28,000 3 second audio clips demonstrating the five disfluencies (Interjection, Prolongation, Block, Sound Repetition and Word Repetition) along with a sixth class that represents fluent speech (no disfluency present). The dataset derives the audio sample from podcast episodes, with manual labels provided by three human annotators.

Preprocessing. The dataset was filtered to keep only those samples which corresponded to a unanimous agreement between all three annotators on the disfluency type. This ensured quality data that had no label dispute.

Training. To detect the presence of each disfluency, five separate CNN models are trained to perform binary classification upon the embedding vector produced by wav2vec for each clip. Each CNN focuses on detecting one of the five types of disfluencies, trained to perform binary classification between fluent speech and the disfluency of interest.

4.3 Facial Emotion

Dataset. Facial emotion detection framework uses two specialized datasets for face detection and emotion classification. The WIDER FACE dataset [21] was used for face detection due to its wide coverage of different conditions, including different poses, scales, and lighting scenarios. For emotion recognition, the AffectNet dataset [22] was used which contains over 1 million labelled face images depicting a range of emotional expressions, including anger, joy, sadness, and surprise. Its richness and diversity enable a nuanced understanding of emotional expressions, thereby improving the model's ability to accurately classify emotions in different situations.

Preprocessing. To prepare the AffectNet dataset for training, the data was preprocessed to focus on relevant features. From each video frame, only the head region was extracted and retained, removing irrelevant background and noise. This cropping ensures that the model concentrates solely on facial features critical for emotion recognition. The WIDER FACE dataset was also refined to enhance the face detection model's performance, with annotations ensuring accurate bounding box generation for a wide variety of face types, sizes, and environments.

Training. The training phase involved fine-tuning YOLOv11 models for both face detection and emotion classification. For face detection, the model was trained over 100 epochs to enhance its ability to localize faces across diverse poses, scales, and lighting

conditions. Similarly, for emotion detection, the YOLOv11 model was fine-tuned for 100 epochs to classify emotions into predefined categories: Anger, Contempt, Disgust, Fear, Happy, Neutral, Sad, and Surprise. During inference, a confidence score threshold of 0.6 was applied to ensure the reliability of emotion predictions. The training process, optimized with extensive data augmentation, resulted in a robust system capable of accurate face localization and emotion recognition in complex real-world scenarios.

4.4 Hand Gesture & Posture

Head Pose Detection. The system uses facial landmarks extracted from MediaPipe Face Mesh [18] to identify key points such as the nose tip, eye corners, and chin. These landmarks are analysed in both 2D and 3D to determine the orientation of the head. The head's direction is classified based on rotation angles: looking left, right, up, or down, or looking forward if the head is neutral. This is achieved by calculating the angles of rotation using perspective-n-point (PnP) techniques.

Hand Gesture Detection. Using landmarks from MediaPipe Pose, the system determines the positions of the wrists relative to the shoulders. If both wrists are close to the shoulders, the hands are classified as folded. If the wrists are outside the video frame, the hands are marked as not visible. In all other cases, the hands are identified as visible.

Posture Detection. The system evaluates the alignment of the shoulders to determine whether the person is standing straight. If the vertical alignment of the shoulders is nearly equal, the person is classified as standing straight. Significant misalignment indicates a slouched or uneven posture.

4.5 Speech Sentiment

Dataset. Emotional TTS Dataset [23] which is an Indian accent speech dataset which consists of 30 minutes of recording for each emotion. It consists of 300-500 audio files for each emotion with each audio file ranging around 3-9 seconds. The Emotion under consideration includes Anger, Calm, Excited, Fear, Happy, Sad.

As the model was trained on a demographic of Indian speakers, there is a potential for cultural and linguistic bias in the responses of the model. The model's effectiveness in classifying the emotion of the speaker through speech may be higher for speakers with an Indian accent than non-Indian speakers .

Training. The Speech Emotion Recognition model was implemented using a BiLSTM architecture and trained to classify six emotions: Sad, Happy, Fear, Excited, Calm, and Anger. Speech features like mfccs, chroma, tonnetz, contrast, pitch, jitter, shimmer and intensity were extracted from the audio. The training process spanned over 10 epochs with early stopping with a patience of 5 epochs and an Adam optimizer with a learning rate of 0.001. At the end of each epoch, accuracy and loss metrics were evaluated to

ensure consistency throughout the length of the training process. Stress levels are classified on the basis of whether the value of speech features exceeds the value of pitch, jitter, shimmer, intensity. The nervousness score was calculated by summing up the values of the jitter and shimmer.

4.6 Text Sentiment

Dataset. The text sentiment model was trained on a Twitter dataset that contains sentences from tweets [24]. It consists of labelled instances expressing various emotions, where each entry contains a sentence conveying an emotional expression and a numerical label representing the emotional category. The labels correspond to the six different emotions: Anger, Surprise, Fear, Love, Joy, Sadness.

Training. The text emotion module fine-tunes RoBERTa to classify six emotions (sadness, joy, love, anger, fear, and surprise) from speech transcripts generated using OpenAI's Whisper model. Tokenized inputs, with a maximum length of 128 tokens, are processed in batches of 64 for efficiency. The AdamW optimizer, set with a learning rate of 5e-5, ensured effective weight adjustments during training. The model is trained over three epochs. Cross-entropy loss is calculated at each step, guiding backpropagation to optimize model performance. This training pipeline is designed to process Whisper-generated speech transcripts, enabling accurate emotion detection for real-world applications.

5 Results

5.1 Grammar Error Correction.

Grammar Error Sentence Detector. The evaluation of the Error Detector [13] was performed by dividing the data into training and test sets using an 80-20 ratio. The model was assessed on multiple metrics including accuracy, precision, recall and F1-score (Table 1). It achieved an accuracy of 0.78, demonstrating the model's ability to distinguish between correct and incorrect sentences with minimum false positives and false negatives.

Grammar Error Corrector. The grammar corrector was evaluated on the following datasets: W&I+LOCNESS [25], JFLEG [26], PIE_Synthetic [27] and C4-200M (held out data). To contrast performance for all possible errors against that for the errors of interest, three of the datasets were filtered. The filtered datasets contain sentences with only the ten targeted grammatical errors, while the unfiltered datasets included other errors too. As expected, the results illustrated that the model performed better on filtered datasets, where error types were controlled, thus achieving higher BLEU and ROUGE scores of 0.89 and 0.95. In contrast, the performance on unfiltered datasets was slightly lower, BLEU score 0.75 and ROUGE score of 0.89 (Table 2).

To further ensure the robustness of our approach, we conducted k-fold cross-validation on the PIE unfiltered dataset. The model consistently achieved strong results across folds, with BLEU score of 0.75 (Table2).

Table 1. Performance metrics of grammar error sentence detector.

Class	**Precision**	**Recall**	**F-Score**
Incorrect	0.84	0.69	0.76
Correct	0.84	0.84	0.84
Accuracy			0.78
Macro Avg.	0.79	0.78	0.78
Weighted Avg.	0.79	0.78	0.78

Table 2. Performance metrics of the grammar corrector on filtered & unfiltered datasets.

Dataset	**BLEU**	**Rouge1**	**Rouge2**	**RougeL**	**RougeL-Sum**
W&I+LOCNESS (unfiltered)	0.75	0.90	0.81	0.89	0.89
JFLEG (unfiltered)	0.64	0.85	0.71	0.84	0.84
PIE_Synthetic (unfiltered)	0.75	0.90	0.88	0.90	0.90
C4-200M (filtered)	0.85	0.93	0.87	0.93	0.93
W&I+LOCNESS (filtered)	0.84	0.92	0.84	0.92	0.92
JFLEG (filtered)	0.89	0.95	0.89	0.95	0.95
PIE_Synthetic (unfiltered) (K-fold Cross Validation)	0.75	0.90	0.88	0.90	0.90

Table 3. BLEU Scores Comparison Between Proposed Grammar Corrector and Existing Models

		MODEL			
		T5-Base (proposed model)	T5-Base (finetuned)	T5-Small (finetuned)	Custom T5 (pre-trained)
DATASET	C4-200M (unfiltered)	-	0.60	0.56	0.435
	C4-200M (filtered)	**0.85**			
	W&I+LOCNESS (filtered)	**0.84**			
	JFLEG (filtered)	**0.89**			

According to P. Jain et al. [28], various fine-tuned T5-based grammar correction models, including T5-Base, T5-Small, and custom pre-trained variants, were evaluated on the C4-200M dataset. As shown in Table 3, the best BLEU score reported among them was 0.60 for the T5-Base fine-tuned model. In contrast, the proposed T5-Base model, fine-tuned specifically to correct the top 10 most common grammatical error types demonstrates significantly better generalization and performance. This focused fine-tuning approach allows the model to target the most frequent and impactful grammar issues in real-world texts, leading to improved correction quality across different contexts. Our model achieves BLEU scores of 0.85, 0.84, and 0.89 on the C4-200M

(Filtered), W&I+LOCNESS (Filtered), and JFLEG (Filtered) datasets, respectively. This improvement indicates not only a more effective training strategy but also a broader adaptability to various linguistic styles and error distributions.

5.2 Speech Disfluency Detection.

As a binary classification task, the evaluation of disfluency detection in the form of accuracy yielded the following results, with the Interjection model having an accuracy of 0.74 trained over 11 epochs (Table 4). Coming to Block and Prolongation, both achieved accuracies of 0.76 over 13 epochs. Finally, the Word Repetition and Sound Repetition models achieved accuracies of 0.75 and 0.82 respectively over 20 epochs.

Table 4. Model metrics for speech disfluency detection.

Model	Accuracy	No of Training Epochs
Interjection	0.74	11
Block	0.76	13
Prolongation	0.76	13
Word Repetition	0.75	20
Sound Repetition	0.82	20

5.3 Face Emotion Recognition.

The face emotion recognition model has 2 modules, both of which were evaluated using confusion matrices to analyse ground truth versus model predictions.

Face Emotion Model. Evaluation of the face emotion model presented a true positive value of 0.76 for Anger, 0.72 for Contempt, 0.70 for Disgust, 0.75 for Fear, 0.86 for Happy, 0.67 for Neutral, 0.69 for Sad and 0.75 for Surprise (Table 5).

Face Detection Model. The evaluation of the face detection model is represented in Table 6 as a confusion matrix. Evaluation yielded a true positive value of 0.93 and a false positive value of 0.07.

Table 5. Confusion matrix for facial emotion recognition.

		TRUE								
		Anger	**Contempt**	**Disgust**	**Fear**	**Happy**	**Neutral**	**Sad**	**Surprise**	**Background**
PREDICTED	**Anger**	0.76	0.03	0.10	0.03	0.00	0.08	0.06	0.02	0.14
	Contempt	0.03	0.72	0.03	0.00	0.06	0.07	0.02	0.01	0.13
	Disgust	0.08	0.03	0.70	0.03	0.03	0.03	0.07	0.02	0.17
	Fear	0.01	0.00	0.02	0.75	0.00	0.01	0.03	0.11	0.09
	Happy	0.00	0.07	0.03	0.01	0.86	0.01	0.01	0.01	0.05
	Neutral	0.05	0.11	0.02	0.02	0.00	0.67	0.09	0.05	0.10
	Sad	0.04	0.03	0.06	0.04	0.01	0.07	0.69	0.02	0.14
	Surprise	0.03	0.02	0.04	0.13	0.03	0.08	0.03	0.75	0.17

Table 6. Confusion matrix for face detection.

		True	
		Face	**Background**
Predicted	**Face**	0.93	1.00
	Background	0.07	0.00

5.4 Speech Emotion Recognition.

The evaluation of speech emotion recognition model was performed by dividing the dataset in an 80-20 train-test split. The model was assessed based on accuracy. The results demonstrate how the threshold, which sets the minimum confidence score for emotion classification, impacts model accuracy. At a 30% threshold, the accuracy was 88.09%, while lowering it to 10% increased accuracy to 94.04% and the actual predicted accuracy is 83.62% (Table 7). Adjusting the threshold helps fine-tune predictions to align better with actual outcomes.

Table 7. Model metrics for speech emotion.

Percentage Threshold	**Accuracy**
Actual Accuracy (without Threshold)	83.62%
30%	88.09%
10%	94.04%

5.5 Text Emotion Recognition

The Text emotion recognition model was assessed based on precision, recall, F1-Score and overall accuracy (Table 8). An overall accuracy of 0.94 was achieved with the macro average of all the emotions being 0.94, 0.87, 0.89 for precision, recall and F1-Score respectively. The weighted average accuracies for all the emotions are 0.94, 0.94, 0.94 for the respective metrics.

Table 8. Model metrics for text emotion recognition.

	Precision	**Recall**	**F1-Score**
Sadness	0.98	0.97	0.98
Joy	0.91	1.00	0.96
Love	1.00	0.68	0.81
Anger	0.93	0.98	0.96
Fear	0.92	0.91	0.91
Surprise	0.86	0.67	0.75
Accuracy			0.94
Macro Avg.	0.94	0.87	0.89
Weighted Avg.	0.94	0.94	0.94

Overall, SpeakSmart has demonstrated satisfactory results and performs reliably, meeting the objectives outlined in the research study. Compared to previously published works, our approach offers higher accuracy and robustness through several enhancements and methodological improvements. The findings of Hao and Hao [10] detected 25 grammatical errors with F-scores of 0.617 and 0.609 for beam sizes of 12 and 1, respectively. Our grammatical correction approach achieved an F-score of 0.78. Lim et al. [1] conducted a study on emotion recognition using eye-tracking technology, resulting in an accuracy of 90%, while our face detection approach yielded an accuracy of 93%. Mustaqeem et al. [2] proposed a framework utilizing radial basis function networks, achieving accuracies of 85.57% on EMO-DB and 77.02% on RVDESS. Our speech sentiment analysis model achieved an accuracy of 83.62%. These results highlight the model's potential as a reliable personal oratory coach.

SpeakSmart provides user feedback in a comprehensive graphical format, making it easy for users to self-evaluate and understand their performance. The 'Result' section for each uploaded video presents multiple insights. For grammar analysis, users can view their original speech alongside a corrected version, accompanied by a pie chart depicting the frequency and occurrence of grammatical errors. Speech disfluencies are highlighted with a dedicated pie chart and a bar graph showing the clip-wise distribution of these disfluencies. Posture alignment and hand gesture attributes are summarized through respective pie charts, offering a clear visual representation of physical behaviour. Additionally, the emotion analysis section features pie charts for speech emotion, textual emotion, and facial emotion frequencies, enabling users to assess their emotional expressions effectively.

6 Conclusions and Future Work

The proposed tool allows an insightful perspective towards communication skills in general which is achieved by analysing the body language, speech patterns, posture and grammar. Advanced sentiment analysis of face, speech and text, along with body language, disfluencies and grammar provide a multimodal statistical view on how well-received a speech delivery may be. This tool thus can be used to supplement the oratory training for students as well as professionals.

SpeakSmart is currently compatible with the English language only, leading to a scope for diversifying the application by expanding the modules to be compatible and work seamlessly with other foreign languages to ensure a more inclusive experience for users across the globe.

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